Examiner only

2. *(a)* (i) Define the *work function* of a material. [1]

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(ii) When a potassium surface is irradiated with light of frequency 7.4×10^{14} Hz, electrons of maximum kinetic energy 1.2×10^{-19} J are ejected at a rate of 2.0×10^{15} electrons per second.

I. Explain, in terms of photons, how, if at all, the maximum kinetic energy of the ejected electrons **and** their rate of ejection would change if a more intense light of the same frequency were used. [3]

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II. Determine whether or not electrons would be ejected from a potassium surface by light of frequency 5.1×10^{14} Hz. Give your reasoning. [4]

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Examiner only

(b) A beam of monochromatic light of wavelength, λ and power, P strikes an absorbing surface normally.

(i) Derive an expression for the number of photons, N, striking the surface per second, in terms of P, λ, h and c. [2]

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(ii) Hence derive an expression for the momentum change per second of the light when it strikes the surface. [2]

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(iii) A student suggests that the answer to (ii) gives the *pressure* that the light exerts on the surface. What *should* she have said instead of *pressure*? [1]

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